

Question	Answer	Marks
7(a)	• force per unit length • force per unit current • length / current perpendicular to field <i>1 mark for any two points, 2 marks for all three points</i>	B2
7(b)	concentric circles around the wire (at least two circles needed)	B1
	spacing between circles increases with distance from wire (at least four circles needed)	B1
	arrows showing direction of field is clockwise	B1
7(c)(i)	(each) wire sits in the (magnetic) field created by the other	B1
	current (in one wire) is perpendicular to (magnetic) field (due to other wire) so (magnetic) force acts (on wire)	B1
7(c)(ii)	arrow drawn, starting from X and pointing towards Y, labelled F	B1
7(c)(iii)	(forces have) equal magnitudes	B1
	(forces are in) opposite directions	B1
7(c)(iv)	no change (in the direction of the force) since both the current in X and the field due to Y have reversed	B1

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